

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 2**Date:** 1/15/24**Course:** Calculus Beta

Use l'Hôpital's Rule to find the limit.

$$\lim_{x \rightarrow 0} \frac{10x^2}{\cos(x) - 1}$$

Substituting $x = 0$ returns the indeterminate form $\frac{0}{0}$. L'Hôpital's Rule states that if $f(a) = g(a) = 0$, f and g are differentiable on an open interval I containing a , and $g'(x) \neq 0$ on I if $x \neq a$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, assuming that the limit on the right exists. This process may be repeated until one no longer has an indeterminate form.

In this case, $f(x) = 10x^2$. Find $f'(x)$.

$$f'(x) = 20x$$

Now find $g'(x)$ where $g(x) = \cos(x) - 1$.

$$g'(x) = -\sin(x)$$

By direct substitution, $\lim_{x \rightarrow 0} -\frac{20x}{\sin(x)}$ still returns the indeterminate form $\frac{0}{0}$ because as x approaches 0, the numerator and denominator of the limit approach zero. Apply l'Hôpital's Rule again and take the limit. Begin by taking the derivatives of both the numerator and the denominator.

$$\begin{aligned} \lim_{x \rightarrow 0} -\frac{20x}{\sin(x)} &= \lim_{x \rightarrow 0} -\frac{\frac{d}{dx}(20x)}{\frac{d}{dx}(\sin(x))} \\ &= \lim_{x \rightarrow 0} -\frac{20}{\cos(x)} \end{aligned}$$

Simplify the expression and take the limit by direct substitution.

$$\begin{aligned} \lim_{x \rightarrow 0} -\frac{20}{\cos(x)} &= -\frac{20}{\cos(0)} \\ &= -20 \end{aligned}$$

Thus, $\lim_{x \rightarrow 0} \frac{10x^2}{\cos(x) - 1} = -20$.